

2.1 (a)

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

Sample mean: $\bar{x} = 1$, $\bar{y} = 2$.

$$(b) \quad b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8$$

$$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 \cdot 1 = 1.2$$

$$(c) \quad \sum_{i=1}^5 x_i^2 = 9 + 4 + 1 + 1 + 0 = 15$$

$$\sum_{i=1}^5 x_i y_i = 12 + 4 + 3 - 1 + 0 = 18$$

$$\sum (x_i - \bar{x})^2 = 10$$

$$\sum x_i^2 - N \bar{x}^2 = 15 - 5 \cdot 1^2 = 10$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$$

$$\sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8$$

$$\left. \begin{array}{l} \sum (x_i - \bar{x})^2 = 10 \\ \sum x_i^2 - N \bar{x}^2 = 15 - 5 \cdot 1^2 = 10 \end{array} \right\} \Rightarrow \sum (x_i - \bar{x})^2 = \sum x_i^2 - N \bar{x}^2$$

$$\left. \begin{array}{l} \sum (x_i - \bar{x})(y_i - \bar{y}) = 8 \\ \sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8 \end{array} \right\} \Rightarrow \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N \bar{x} \bar{y}$$

(d)

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$\hat{y}_i = b_1 + b_2 x_i = 1.2 + 0.8 x_i, \text{ and } \hat{e}_i = y_i - \hat{y}_i$$

$$\text{Sample variance of } y: S_y^2 = \sum_{i=1}^N \frac{(y_i - \bar{y})^2}{N-1} = \frac{1}{4} (4 + 1 + 0 + 1 + 4) = 2.5$$

$$\text{Sample variance of } x: S_x^2 = \sum_{i=1}^N \frac{(x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2.5$$

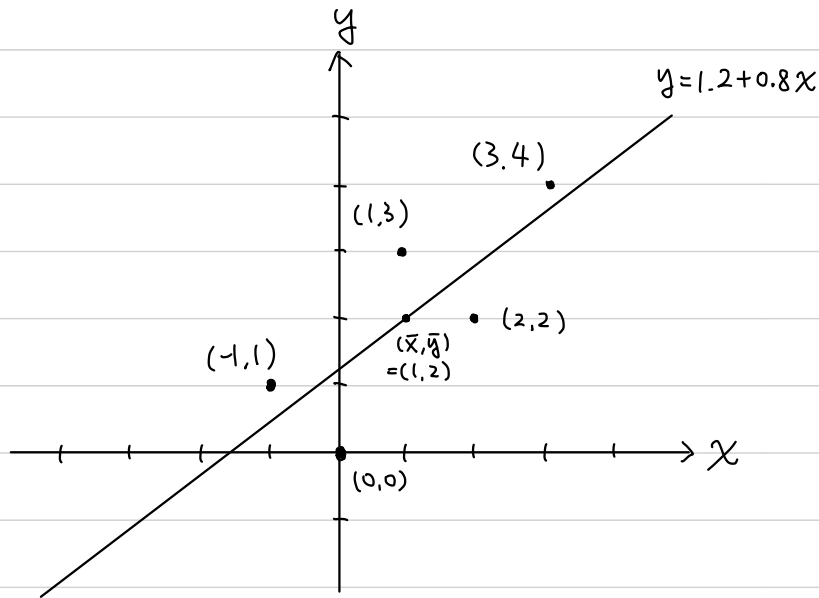
Sample variance between X and y : $S_{xy} = \sum_{i=1}^N \frac{(y_i - \bar{y})(x_i - \bar{x})}{N-1} = \frac{8}{4} = 2$

Sample correlation between X and y : $r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{2.5} = 0.8$

Coefficient of variation of X : $CV_x = 100 \cdot \left(\frac{S_x}{\bar{x}} \right) = 100 \cdot \sqrt{2.5} = 25\sqrt{5}$

Median of x : 1.

(e)



x	y
3	4
2	2
1	3
-1	1
0	0

(g) $\bar{y} = 2$, $b_1 + b_2 \bar{x} = 1.2 + 0.8 \cdot 1 = 2 \Rightarrow \bar{y} = b_1 + b_2 \bar{x}$

(h) $\bar{\hat{y}} = \sum \frac{\hat{y}_i}{N} = \frac{1}{5} \cdot 10 = 2 = \bar{y}$

(i) $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

(j) $\hat{\text{var}}(b_2|x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$\text{se}(b_2) = \sqrt{\hat{\text{var}}(b_2|x)} = \sqrt{0.12} = \frac{\sqrt{3}}{5}$

2.22

```
a. > library(POESRdata)
> mod1 <- lm(totalscore ~ small, data = star5_small)
> smod1 <- summary(mod1)
> smod1

Call:
lm(formula = totalscore ~ small, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-191.63  -50.63   -7.12   42.37  324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  918.628     2.536  362.260  <2e-16 ***
small         9.989      4.611   2.166  0.0305 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.36 on 1198 degrees of freedom
Multiple R-squared:  0.003903,    Adjusted R-squared:  0.003071
F-statistic: 4.694 on 1 and 1198 DF,  p-value: 0.03047
```

The estimated coefficient on small (9.989) indicates that student in small classes score about 10 points higher on average than students in regular classes, and this difference is statistically significant at the 5% level ($p=0.0305$).

```
b. > mod2 <- lm(readscore ~ small, data = star5_small)
> smod2 <- summary(mod2)
> smod2

Call:
lm(formula = readscore ~ small, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-76.123 -21.123  -3.356  15.877  192.877

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  434.123     1.044  416.021  < 2e-16 ***
small         5.466      1.897   2.881  0.00403 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 1198 degrees of freedom
Multiple R-squared:  0.006882,    Adjusted R-squared:  0.006053
F-statistic: 8.301 on 1 and 1198 DF,  p-value: 0.004032
```

The estimated coefficient on small (5.466) indicates that student in small classes score about 5.5 points higher in reading on average than students in regular classes, and this difference is statistically significant at the 1% level ($p=0.004$).

```
> mod3 <- lm(mathscore ~ small, data = star5_small)
> smod3 <- summary(mod3)
> smod3

Call:
lm(formula = mathscore ~ small, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-145.505  -35.028   -6.505   28.495  141.495

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  484.505     1.677  288.884  <2e-16 ***
small         4.522      3.049   1.483   0.138
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.52 on 1198 degrees of freedom
Multiple R-squared:  0.001832,    Adjusted R-squared:  0.0009992
F-statistic: 2.199 on 1 and 1198 DF,  p-value: 0.1383
```

The estimated coefficient on small (4.522) indicates that student in small classes score about 4.5 points higher in math on average than students in regular classes, but this difference is NOT statistically significant.

The estimates suggest that students in small classes perform better in both reading and math. However, the effect is statistically significant for reading scores but not for math scores.

C.

```
> mod4 <- lm(totalscore ~ aide, data = star5_small)
> smod4 <- summary(mod4)
> smod4

Call:
lm(formula = totalscore ~ aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-195.14  -52.14   -6.95   40.95  330.86

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  922.145     2.640  349.250  <2e-16 ***
aide        -1.396      4.437   -0.315    0.753
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.5 on 1198 degrees of freedom
Multiple R-squared:  8.267e-05, Adjusted R-squared:  -0.000752
F-statistic: 0.09904 on 1 and 1198 DF, p-value: 0.753
```

The estimated coefficient on aide (-1.396) indicates that student in a regular classes with a teacher aide score about 1.4 points lower on average than a class without a teacher aide, but this difference is NOT statistically significant.

d.

```
> mod5 <- lm(readscore ~ aide, data = star5_small)
> smod5 <- summary(mod5)
> smod5

Call:
lm(formula = readscore ~ aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-77.908  -21.536   -2.908   15.464  191.092

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  435.9084     1.0882  400.585  <2e-16 ***
aide        -0.3719     1.8285   -0.203    0.839
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.29 on 1198 degrees of freedom
Multiple R-squared:  3.453e-05, Adjusted R-squared:  -0.0008002
F-statistic: 0.04137 on 1 and 1198 DF, p-value: 0.8389
```

The estimated coefficient on aide (-0.3719) indicates that student in a regular classes with a teacher aide score about 0.4 point lower on average in reading than a regular class without a teacher aide, but this difference is NOT statistically significant.

```
> mod6 <- lm(mathscore ~ aide, data = star5_small)
> smod6 <- summary(mod6)
> smod6

Call:
lm(formula = mathscore ~ aide, data = star5_small)

Residuals:
    Min       1Q   Median       3Q      Max
-147.236  -32.236   -7.212   27.788  140.788

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  486.236     1.744  278.731  <2e-16 ***
aide        -1.024      2.931   -0.349    0.727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

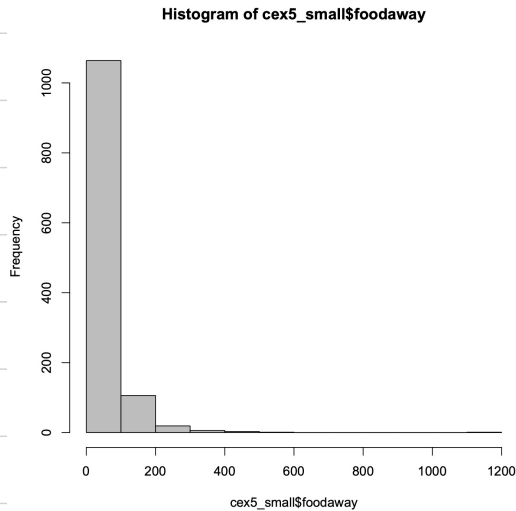
Residual standard error: 48.56 on 1198 degrees of freedom
Multiple R-squared:  0.0001019, Adjusted R-squared:  -0.0007327
F-statistic: 0.1221 on 1 and 1198 DF, p-value: 0.7268
```

The estimated coefficient on aide (-1.024) indicates that student in a regular class with a teacher aide score about 1 point lower on average in math than a regular class without a teacher aide, but this difference is NOT statistically significant.

The estimates suggest that students in a regular class without a teacher aide perform better than a regular class with a teacher aide in both reading and math, but both of them are not statistically significant.

2.25

a.



← Histogram of FOODAWAY

↓ Summary statistics

```
> library(POE5Rdata)
> data("cex5_small")
> hist(cex5_small$foodaway, col='grey')
> summary(cex5_small$foodaway)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00

mean: 49.27 median: 32.55

25th percentiles: 12.04

75th percentiles: 67.5

b.

```
> summary(cex5_small$foodaway[cex5_small$college==1])
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00 12.04 32.55 49.27 67.50 1179.00

> summary(cex5_small$foodaway[cex5_small$advanced==1])
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00 21.67 48.15 73.15 90.00 1179.00

> summary(cex5_small$foodaway[cex5_small$college==0 & cex5_small$advanced==0])
Min. 1st Qu. Median Mean 3rd Qu. Max.
-Inf -Inf -Inf -Inf -Inf -Inf

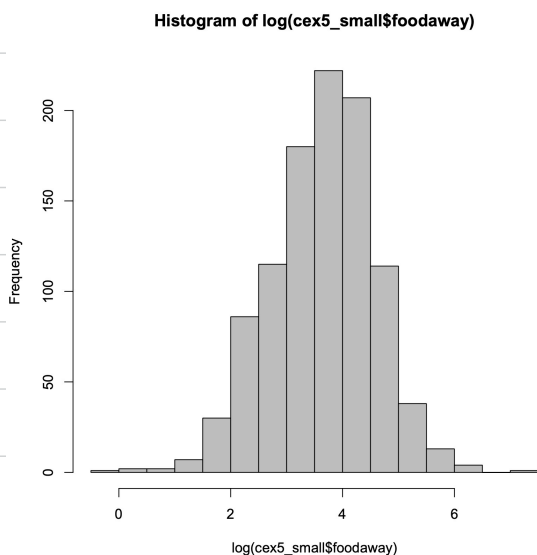
> table(cex5_small$college, cex5_small$advanced)
```

	0	1
0	943	257

	with an advanced degree	with a college degree member
mean	73.15	49.27
median	48.15	32.55

Since there are no data for households with no advanced or college degree member, it doesn't have mean nor median values.

c.



```
> summary(log(cex5_small$foodaway))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-Inf	2.488	3.483	-Inf	4.212	7.072

Since at least one data in FOODAWAY is 0, and the log of 0 doesn't exist, so $\ln(\text{FOODAWAY})$ has less observation than FOODAWAY.

d.

```
> mod1 <- lm(log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway > 0, ])
> smod1 <- summary(mod1)
> smod1
```

Call:

```
lm(formula = log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway > 0, ])
```

Residuals:

Min	1Q	Median	3Q	Max
-3.6547	-0.5777	0.0530	0.5937	2.7000

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1293004	0.0565503	55.34	<2e-16 ***
income	0.0069017	0.0006546	10.54	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom

Multiple R-squared: 0.09826, Adjusted R-squared: 0.09738

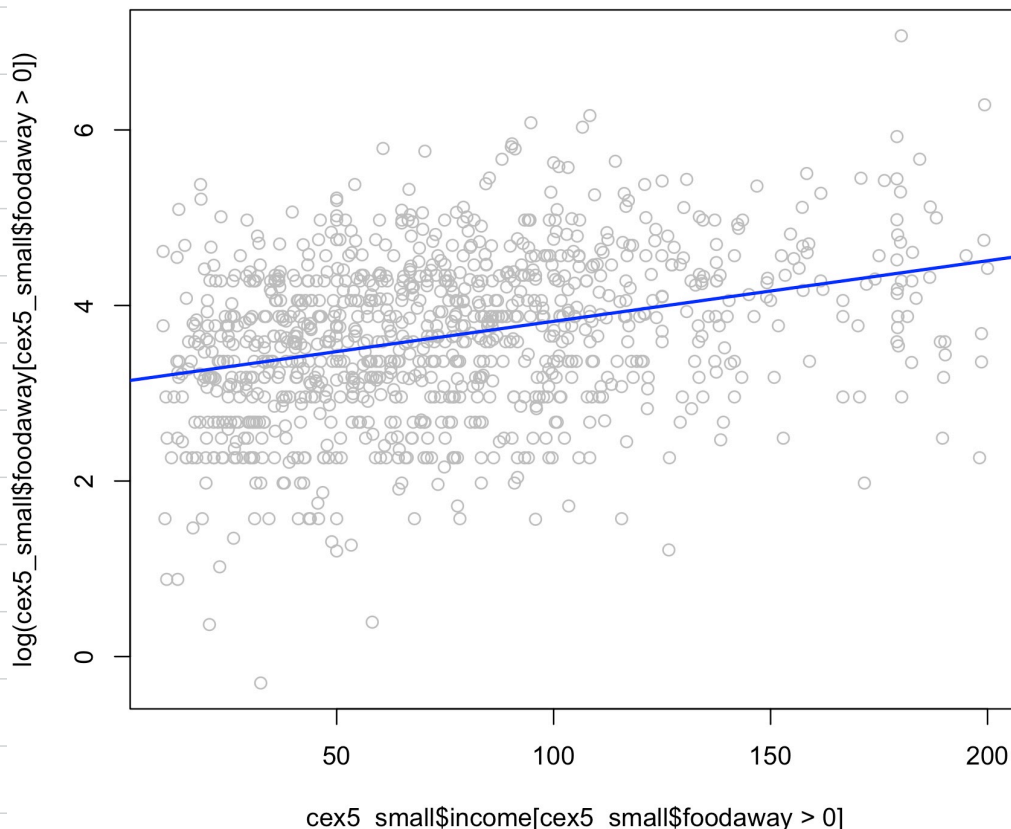
F-statistic: 111.1 on 1 and 1020 DF, p-value: < 2.2e-16

Estimate of the linear regression:

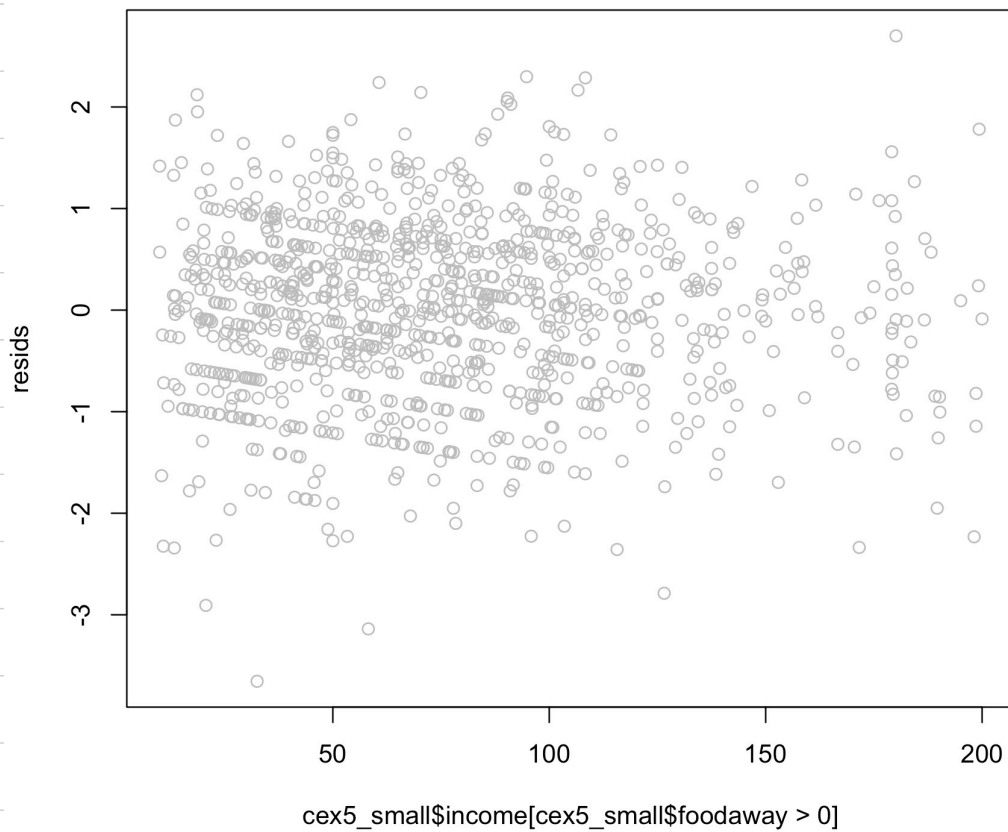
$$\ln(\text{FOODAWAY}) = 3.1293004 + 0.0069017 \cdot \text{INCOME} + e$$

The estimate slope 0.0069017 means that if the monthly income goes up by \$100, the log of the expected food away from home expenditure per month per person will increase approximately by 0.0069017.

e.



f.



We can see that the residuals are not completely random:
As the income increases, the variance of the residual increases.